Exercises

Exercise 1

Assume we have a dataset $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)})\} = \{(1,2), (2,4), (3,5)\}$ of points in the 2D plane, and we want to estimate the following linear regression model:

$$\hat{y} = w_1 x + w_0.$$

Compute w_0, w_1 and $\hat{y}^{(1)}$, $\hat{y}^{(2)}$, $\hat{y}^{(3)}$.

Solution:

- we have the following formulas:

$$\boldsymbol{X} = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} & \dots & x_d^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} & \dots & x_d^{(2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_1^{(n)} & x_2^{(n)} & \dots & x_d^{(n)} \end{bmatrix} = \begin{bmatrix} \boldsymbol{x}^{(1)\top} \\ \boldsymbol{x}^{(2)\top} \\ \vdots \\ \boldsymbol{x}^{(n)\top} \end{bmatrix} \in \mathbb{R}^{n \times (d+1)},$$

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$$\boldsymbol{y} = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(n)} \end{bmatrix} \in \mathbb{R}^{n \times 1},$$

$$\boldsymbol{w}^* = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y} \in \mathbb{R}^{(d+1) \times 1},$$

$$\hat{\boldsymbol{y}} = \begin{bmatrix} \hat{y}^{(1)} \\ \hat{y}^{(2)} \\ \vdots \\ \hat{y}^{(n)} \end{bmatrix} \in \mathbb{R}^{n \times 1},$$

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{w} \in \mathbb{R}^{n \times 1}.$$

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- in our case, $n = 3$, $d = 1$, so we apply the formulas:

$$\boldsymbol{X} = \begin{bmatrix} 1 & x^{(1)} \\ 1 & x^{(2)} \\ 1 & x^{(3)} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \in \mathbb{R}^{3\times 2},$$

$$\boldsymbol{y} = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ y^{(3)} \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} \in \mathbb{R}^{3\times 1},$$

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$$
\begin{aligned}
\boldsymbol{w}^* &= \left(\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} \\
&= \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}^{-1} \begin{bmatrix} 11 \\ 25 \end{bmatrix} \\
&= \begin{bmatrix} \frac{7}{3} & -1 \\ -1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 11 \\ 25 \end{bmatrix} \\
&= \begin{bmatrix} \frac{2}{3} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} \in \mathbb{R}^{2 \times 1},
\end{aligned}
$$

$$
\hat{\boldsymbol{y}} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} \frac{2}{3} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} \frac{13}{6} \\ \frac{22}{6} \\ \frac{31}{6} \end{bmatrix} = \begin{bmatrix} \hat{y}^{(1)} \\ \hat{y}^{(2)} \\ \hat{y}^{(3)} \end{bmatrix} \in \mathbb{R}^{3 \times 1}.
$$